

Exponential functions



1 Consider the function $y = 5 \times 4^x$. Find the value of y when:

a $x = 0$

b $x = 2$

c $x = -1$

d $x = 3$

2 Consider the function $f(x) = \left(\frac{4}{3}\right)^x$.

a Evaluate $f(0)$.

b Evaluate $f(-2)$.

3 Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

4 Find the missing coordinate in each ordered pair that is a solution to $y = 5^x$:

a $(3, 1)$

b $(1, 5)$

c $(-1, 1)$

d $\left(1, \frac{1}{25}\right)$

5 Find the missing coordinate in each ordered pair that is a solution to $y = -2^x$:

a $(5, 1)$

b $\left(1, -\frac{1}{8}\right)$

c $(-1, 1)$

d $(1, -16)$

6 If the point $(3, m)$ lies on the curve $y = 2^x$, solve for m .

7 The points $(3, n)$, $(k, 16)$ and $\left(m, \frac{1}{4}\right)$ all lie on the curve with equation $y = 2^x$.

a Solve for n .

b Solve for k .

c Solve for m .

Characteristics of exponential functions

8 Do either of the functions $y = 9^x$ or $y = 9^{-x}$ have x -intercepts? Explain your answer.

9 For what values of x will 6^x have a negative value?

10 Determine the y -intercept of all exponential functions of the form:

a $y = a^x$

b $y = a^{-x}$

c $y = -a^x$

d $y = -a^{-x}$



11 Consider an exponential curve with equation of the form $y = a^x$, where $a > 0$.

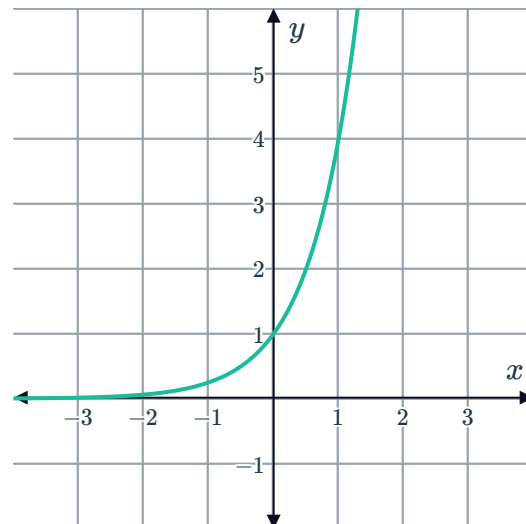
- a** What is the y -value of the y -intercept of all exponential curves of this form?
- b** What is the y -value of the y -intercept of $y = (3.5)^x$?

12 Consider the expression 3^x .

- a** Evaluate the expression when $x = -4$.
- b** Evaluate the expression when $x = 0$.
- c** Evaluate the expression when $x = 4$.
- d** What happens to the value of 3^x as x gets larger?
- e** What happens to the value of 3^x as x gets smaller?

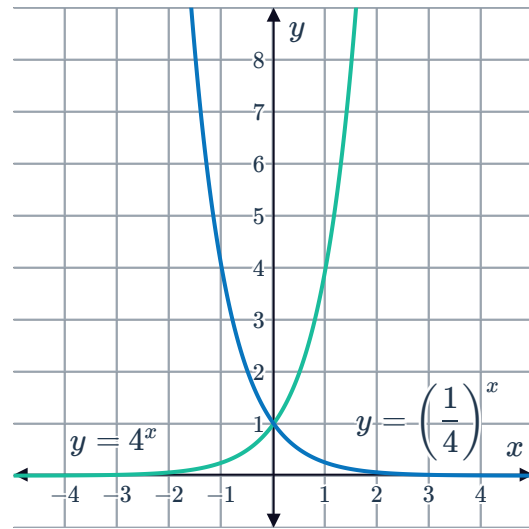
13 Consider the graph of the function $y = 4^x$:

- a** Is each y -value of the function positive or negative?
- b** State the value of y the graph approaches but does not reach.
- c** State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.



14 Given the function $y = -10^x$, what is the largest possible function value? Explain your answer.

- 15 Consider the graphs of the functions $y = 4^x$ and $y = \left(\frac{1}{4}\right)^x$.



- Which function is an increasing function?
- How would you describe the rate of increase of $y = 4^x$?

- 16 Of the two functions $y = 4^x$ and $y = 5^x$, which increases more rapidly for $x > 0$?

- 17 Consider the two functions $y = 4^{-x}$ and $y = 5^{-x}$. Which function decreases more rapidly for $x > 0$?

- 18 Consider the function $y = 3^x$.

- Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

- Describe the behavior of the function as x increases.
- State the domain of the function.
- State the range of the function.

19 Consider the function $y = 2^{-x}$.



a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c Find the y -value of the y -intercept of the function.

d State the domain of the function.

e State the range of the function.

20 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

b Can the value of y ever be zero or positive? Explain your answer.

c Is $y = -(2^x)$ an increasing or decreasing function?

d Describe the behavior of the function as x increases.

e State the domain of the function.

f State the range of the function.

21 Consider the graph of the function $y = 2^x$:

a State the y -intercept of this graph.

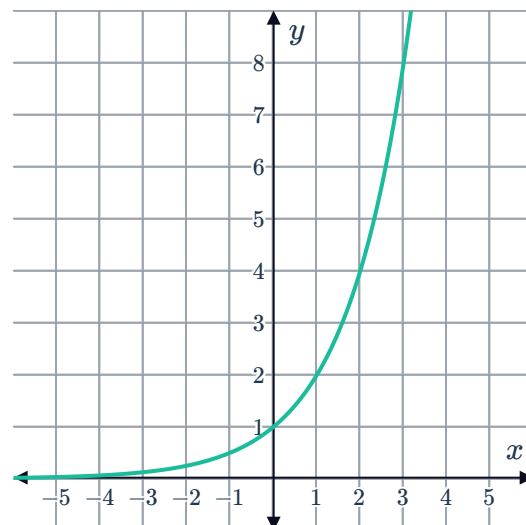
b Does the graph have an x -intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.



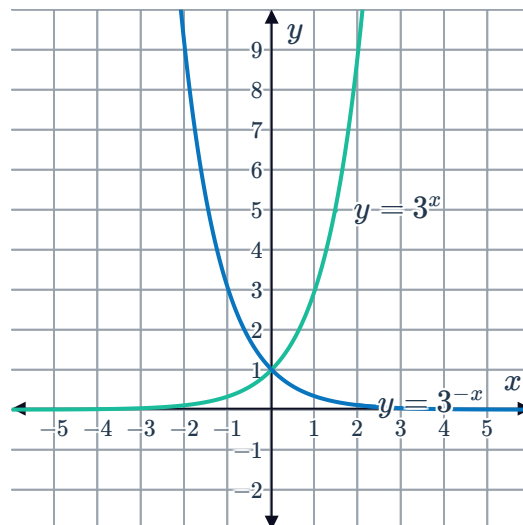
22 Consider the functions $y = 2^{-x}$, $y = 3^{-x}$ and $y = 5^{-x}$.



- State a characteristic that all the graphs of the functions have in common.
- What is the y -value of the shared y -intercept?
- Describe the nature of these functions for large values of x .

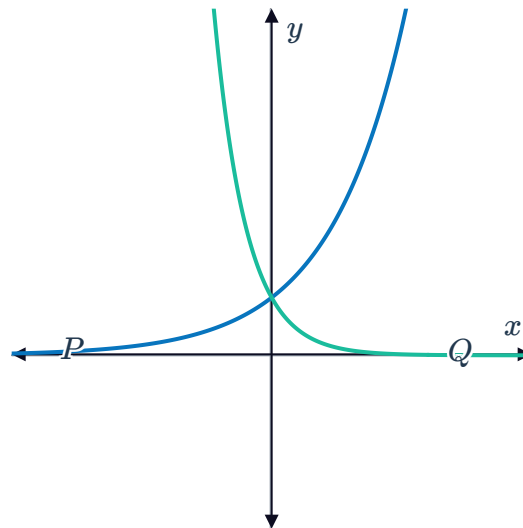
23 Consider the graph of the following functions $y = 3^x$ and $y = 3^{-x}$:

- State the coordinates of the point of intersection of the two curves.
- Describe the behavior of both these functions for large values of x .



24 Consider the given graphs of the two exponential functions P and Q :
State whether the following pairs of equations could be the equations of the graphs P and Q :

- | | |
|------------------------|----------------------------|
| a $P : y = 2^x$ | b $P : y = (3.5)^x$ |
| $Q : y = 2^{-x}$ | $Q : y = 6^{-x}$ |
| c $P : y = 2^x$ | d $P : y = 5^x$ |
| $Q : y = 5^{-x}$ | $Q : y = 2^{-x}$ |

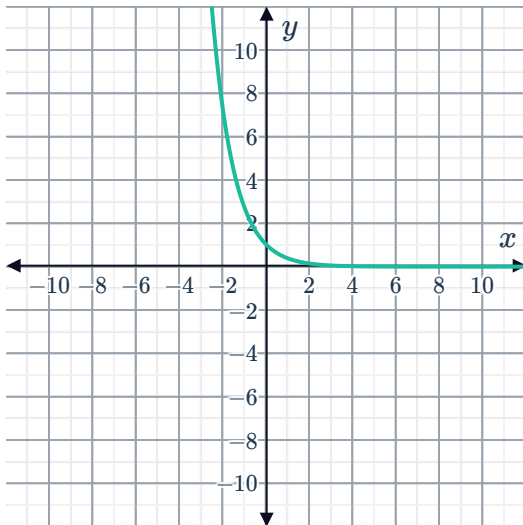




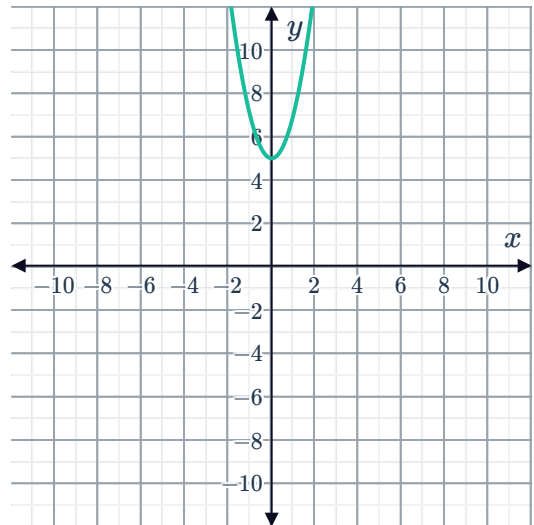
Graphs of exponential functions

25 Determine whether the following graphs show an exponential function:

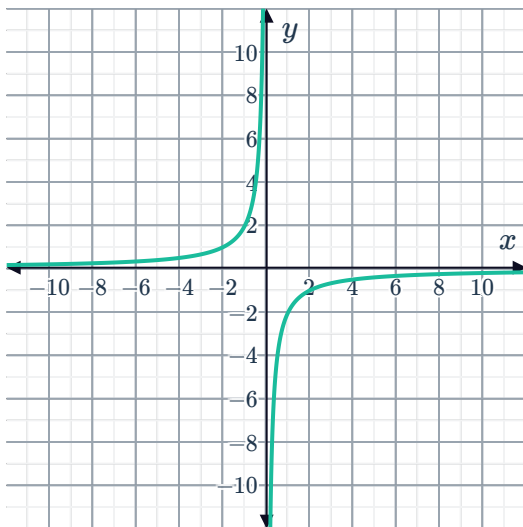
a



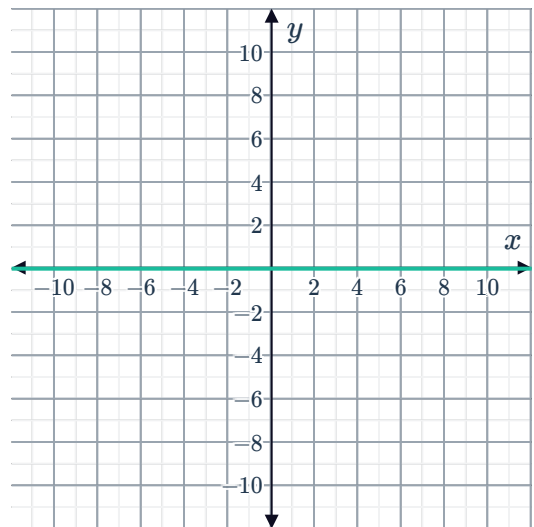
b



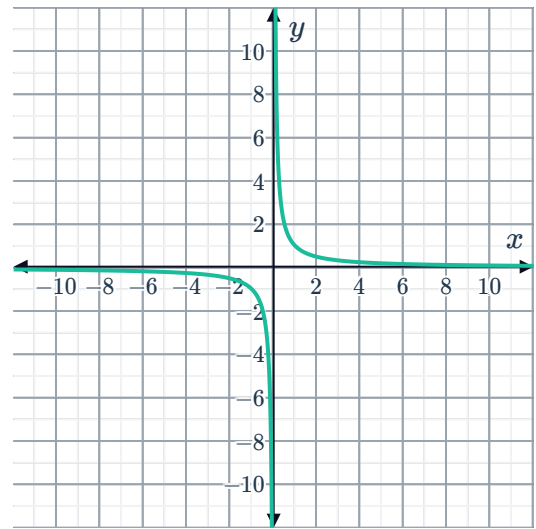
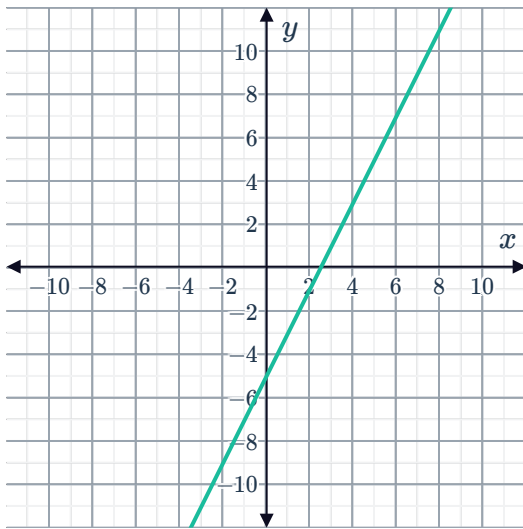
c



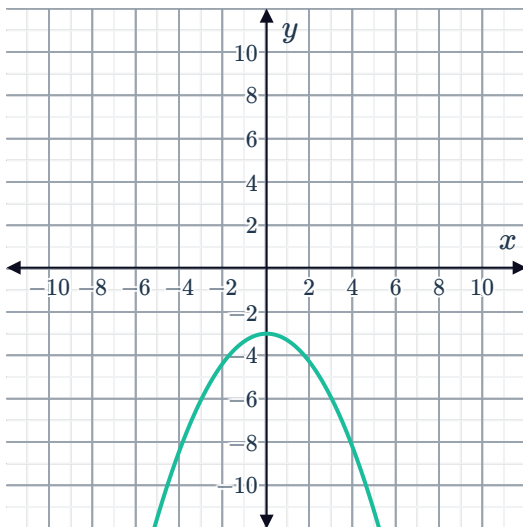
d



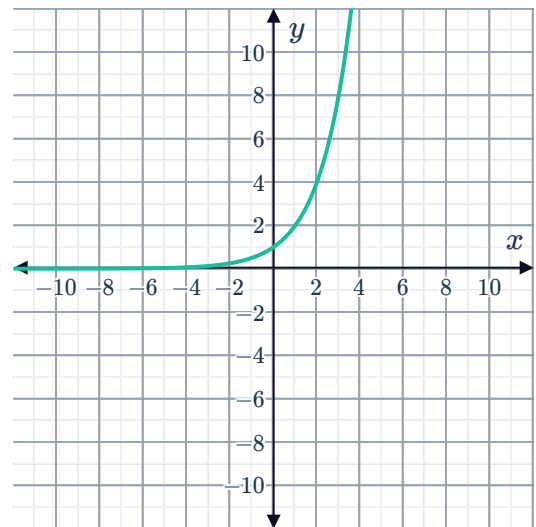
e



g



h



26 Consider the equation $y = 3^x$.

- Complete the table of solutions for the given equation.
- Plot the points C , D , and E on a coordinate plane.
- Sketch the curve that results from the entire set of solutions for the equation being graphed.

Point	Coordinates
A	$(-3, \frac{1}{27})$
B	$(-2, \frac{1}{9})$
C	$(-1, \frac{1}{3})$
D	$(0, 1)$
E	$(1, 3)$
F	$(2, 9)$
G	$(3, 27)$

27 Consider the equation $y = 2^x$.



a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

- b Sketch the graph of the function.
- c What happens to the y -values as x increases?
- d What happens to the y -values as x decreases?
- e Does the curve cross the x -axis?
- f At what value of y does the graph cross the y -axis?

28 Consider the equation $y = 2^{-x}$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

- b Sketch the graph of the function.
- c What happens to the y -values as x increases?
- d What happens to the y -values as x decreases?
- e Does the curve cross the x -axis?
- f At what value of y does the graph cross the y -axis?

29 For each of the following equations:

- i Find the y -intercept of the curve.
- ii Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

- iii Find the horizontal asymptote of the curve.
- iv Sketch the graph of the function.

a $y = 6^x$

b $y = 3^x$

c $y = 10^x$

d $y = 3^{-x}$



30 Consider the function $y = 6^x$.

- a** Can the value of y ever be zero or negative? Explain your answer.
- b** As the values of x get larger and larger, what value does y approach?
- c** As the values of x get smaller and smaller, what value does y approach?
- d** Find the y -value of the y -intercept of the curve.
- e** How many x -intercepts does the curve have?
- f** Sketch the graph of the function.

31 Consider the function $y = 8^{-x}$.

- a** Can the value of y ever be zero or negative? Explain your answer.
- b** As the value of x gets larger and larger, what value does y approach?
- c** As the value of x gets smaller and smaller, what value does y approach?
- d** Find the y -value of the y -intercept of the curve.
- e** How many x -intercepts does the curve have?
- f** Sketch the graph of the function.

32 Consider the function $y = -3^x$.

- a** Find the y -intercept of the curve $y = -3^x$.
- b** Find the horizontal asymptote of the curve $y = -3^x$.
- c** Now, sketch the graph of the function.
- d** Is the function $y = -3^x$, an increasing or decreasing function?

33 Consider the function $y = -6^x$.

- a** Find the y -intercept of the curve $y = -6^x$.
- b** Find the horizontal asymptote of the curve $y = -6^x$.
- c** Now, sketch the graph of the function.
- d** Is the function $y = -6^x$, an increasing or decreasing function?

34 Consider the function $y = -3^{-x}$.



- a Find the y -intercept of the curve $y = -3^{-x}$.
- b Find the horizontal asymptote of the curve $y = -3^{-x}$.
- c Now, sketch the graph of the function.
- d Is the function $y = -3^{-x}$, an increasing or decreasing function?

35 Consider the function $y = \left(\frac{1}{2}\right)^x$.

- a Determine whether the following functions are equivalent to $y = \left(\frac{1}{2}\right)^x$:
 - i $y = \frac{1}{2^x}$
 - ii $y = 2^{-x}$
 - iii $y = -2^x$
 - iv $y = -2^{-x}$
- b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{2}\right)^x$ from the graph of $y = 2^x$.
- c Graph the functions $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ on the same coordinate plane.

36 Consider the function $y = \left(\frac{1}{3}\right)^x$.

- a Rewrite the function in the form $y = k^{-x}$.
- b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{3}\right)^x$ from the graph of $y = 3^x$.
- c Graph the functions $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$ on the same coordinate plane.

37 Consider the exponentials $y = 3^x$ and $y = 2^x$



- Find the equation of the horizontal asymptote of the curve $y = 3^x$.
- Find the equation of the horizontal asymptote of the curve $y = 2^x$.
- Complete the table of values for $y = 3^x$:

x	-3	-2	-1	0	1	2	3
y							

- Complete the table of values for $y = 2^x$:

x	-3	-2	-1	0	1	2	3
y							

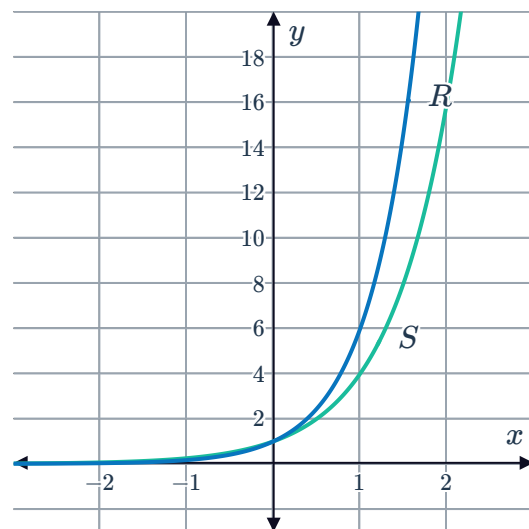
- Now, sketch the graphs of $y = 3^x$ and $y = 2^x$ on the same coordinate plane.
- State the coordinates of the point at which the two graphs intersect.

38 Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

- Sketch the three functions on the same coordinate plane.
- State the y -intercept of each curve.
- Describe the nature of these functions for increasingly large values of x .

39 Consider the graphs of the two exponential functions R and S :

- One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$. Which is the graph of $y = 6^x$?
- For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.



40 Consider the equation $y = -10^x$.



- a Jenny thinks she has found a set of solutions for the equation as shown in the table:

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

She notices that all the y -values are negative and concludes that for any value of x , y must always be negative. Is she correct? Explain your answer.

- b Sketch the graph of $y = -10^x$.
- c Find the values of x for which $y = 0$.